



FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH
HONORS

CSE3433 SOFTWARE ARCHITECTURE

SEMESTER II 2025/2026

Cloud-Native Learning Management System (LMS)

(Cloud-Native / Layered Architecture)

Individual Logbook

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Backend and Database Lead

Week 1

Weekly Progress

- **Undertaken Tasks:**
 - Formed the 3-person project team and held an initial brainstorming session to discuss potential project themes.
 - Researched cloud-native application concepts and potential cloud service providers.
- **Individual Contributions:**
 - Led the discussion on utilizing a microservices or service-oriented architecture style for a web platform.
 - Evaluated the feasibility of building a core system using Java web technologies integrated with modern cloud infrastructure.
- **Challenges:** Selecting a domain that benefits heavily from cloud-native scalability.
- **Action Plans:** Conduct a comparative analysis of system ideas before the next meeting.

GenAI Usage Record

- **Date:** Week 1
- **Tool Used:** Gemini
- **Purpose:** Brainstorming core system domains suitable for cloud-native architectures.
- **Prompt Used:** List 5 university project ideas that benefit from a cloud-native microservices architecture and cloud-hosted databases.
- **Output Summary:** Provided ideas including an e-commerce engine, an LMS and a ride-sharing portal.
- **How It Was Used:** Used to narrow down the team choice to a Cloud-Native Learning Management System (LMS).

Week 2

Weekly Progress

- **Undertaken Tasks:**
 - Finalized the project topic: Cloud-Native Learning Management System (LMS).
 - Outlined the high-level scope of the system and defined individual member responsibilities.
- **Individual Contributions:**
 - Assigned specific sections for Deliverable 1 among team members.
 - Began looking into Aiven.io for managing cloud-hosted database storage.
- **Challenges:** Aligning the timeline with our individual strengths.
- **Action Plans:** Draft the fundamental architectural baseline for the proposal.

GenAI Usage Record

- **Date:** Week 2
- **Tool Used:** Gemini
- **Purpose:** Understanding Aiven.io integration capabilities.
- **Prompt Used:** How can an academic Java application connect to a managed cloud database on Aiven.io safely?
- **Output Summary:** Explained connection pooling, SSL certificate configuration and environment variables.
- **How It Was Used:** Formulated the baseline infrastructure approach for our project scope.

Week 3

Weekly Progress

- **Undertaken Tasks:**

- Began drafting Phase 1 (Deliverable 1) - Project Proposal.
- Aligned system goals with the requirements of a cloud-native framework.
- **Individual Contributions:**
 - Wrote the **Selected Architecture Style** section, justifying the choice of a cloud-native, microservices-driven approach.
 - Formulated the primary **System Objectives** focused on high availability, decoupled storage and seamless student access.
- **Challenges:** Ensuring the architecture chosen explicitly addresses the problems of traditional, monolithic campus legacy apps.
- **Action Plans:** Compile the initial functional requirements into structured use cases.

GenAI Usage Record

- **Date:** Week 3
- **Tool Used:** ChatGPT
- **Purpose:** Structuring architectural justifications.
- **Prompt Used:** What are the key structural benefits of choosing a Cloud-Native Microservices architecture over a monolithic architecture for an educational learning platform?
- **Output Summary:** Highlighted independent service scalability, fault isolation and decoupled data layers.
- **How It Was Used:** Refined the justification text inside the Project Proposal document.

Week 4

Weekly Progress

- **Undertaken Tasks:**
 - Finalized all individual parts of the Project Proposal.
 - Conducted an internal review to ensure cross-functional requirements aligned with the overarching proposal.

- **Individual Contributions:**
 - Drafted the **Initial Functional Requirements**, clearly segmenting core modules like User Authentication, Course Management and Progress Tracking.
 - Collaborated on synthesizing individual team roles into a definitive assignment matrix.
- **Challenges:** Keeping the scope of the initial requirements realistic for a semester-long project.
- **Action Plans:** Format and compile the final PDF for Phase 1 submission.

GenAI Usage Record

- **Date:** Week 4
- **Tool Used:** Gemini
- **Purpose:** Standardizing requirements formatting.
- **Prompt Used:** Draft a structured list of initial functional requirements for an LMS user management and course enrollment module using standard software engineering templates.
- **Output Summary:** Generated formal requirement statements (FR-1, FR-2) with clear inputs, processes and outputs.
- **How It Was Used:** Incorporated directly into the Initial Functional Requirements section of the proposal.

Week 5 (PHASE 1 SUBMISSION)

Weekly Progress

- **Undertaken Tasks:**
 - Reviewed, finalized and **submitted Phase 1 (Deliverable 1): Project Proposal.**
 - Initiated planning for Phase 2 (Architecture Design Document).
- **Individual Contributions:**

- Performed final quality checks on architectural descriptions and system requirements.
- Hosted the post-submission meeting to allocate the architectural diagram tasks.
- **Challenges:** Transitioning theoretical requirements into concrete component structures.
- **Action Plans:** Gather reference templates for cloud-native deployment layouts.

Week 6

Weekly Progress

- **Undertaken Tasks:**
 - Commenced full development of Deliverable 2 (Architecture Design due Week 7).
 - Began defining the non-functional criteria needed for the platform.
- **Individual Contributions:**
 - Documented the comprehensive **Functional Requirements** and **Non-functional Requirements** (focusing on scalability, performance latency and cloud data encryption bounds).
 - Outlined the **Service or Component Identification** section, breaking the system into standalone modular engines.
- **Challenges:** Defining measurable non-functional metrics that are testable in a cloud environment.
- **Action Plans:** Translate the component lists into modeling diagrams.

GenAI Usage Record

- **Date:** Week 6
- **Tool Used:** Gemini
- **Purpose:** Defining cloud-native non-functional metrics.

- **Prompt Used:** What are standard, measurable non-functional requirements for an academic cloud application regarding response times and data security?
- **Output Summary:** Suggested targets like <2 second response times, 99.9% availability and TLS 1.3 encryption constraints.
- **How It Was Used:** Applied these metrics directly to the system's explicit Non-functional Requirements list.

Week 7 (PHASE 2 SUBMISSION)

Weekly Progress

- **Undertaken Tasks:**
 - Drafted, verified and **submitted Phase 2 (Deliverable 2): Architecture Design.**
- **Individual Contributions:**
 - Designed the Interaction Diagram detailing the sequence logic and request-response flow between the client browser, frontend JSP interfaces, Controller Servlets and Data Access Objects (DAO).
 - Designed the Deployment Diagram showing the web application server deployment environment (Apache Tomcat node), runtime execution environments and external secure database hooks to the cloud-hosted Aiven.io database cluster.
- **Challenges:** Ensuring proper visual modeling notation for cloud services inside classic UML structural diagrams.
- **Action Plans:** Set up the GitHub version control environment and begin prototype setup.

GenAI Usage Record

- **Date:** Week 7
- **Tool Used:** Gemini

- **Purpose:** Verifying UML Deployment Diagram architecture conventions for cloud nodes.
- **Prompt Used:** How should an external cloud database cluster like Aiven be represented in a standard UML deployment diagram?
- **Output Summary:** Advised modeling it as an execution environment node with a «cloud service» stereotype containing the database component.
- **How It Was Used:** Guided the layout configuration within our visual modeling software.

Week 8

Weekly Progress

- **Undertaken Tasks:**
 - Created the main team GitHub repository and structured the base workspace directories.
 - Provisioned the live data storage instances on the Aiven.io web console.
- **Individual Contributions:**
 - Configured the connection profiles for the cloud database.
 - Constructed the base database initialization scripts, assuring clean relational schemas for users, courses and materials.
- **Challenges:** Resolving remote connection handshake boundaries and firewall restrictions with cloud endpoints.
- **Action Plans:** Write the core connection pools and prepare for Project Observation 1.

GenAI Usage Record

- **Date:** Week 8
- **Tool Used:** Gemini
- **Purpose:** Generating SQL schema configurations matching a microservices data split.

- **Prompt Used:** Provide a SQL script for user tracking and course tracking tables with foreign key constraints optimized for a modular architecture setup.
- **Output Summary:** Delivered structured DDL SQL commands complete with normalized tables and indexing guidelines.
- **How It Was Used:** Integrated into the initial master database migration script.

Week 9

Weekly Progress

- **Undertaken Tasks:**
 - Demonstrated connectivity validation between the base application framework and the Aiven.io cloud layer to team.
- **Individual Contributions:**
 - Showcased the live database connection routine and walked through how data persists across cloud nodes during verification checks.
 - Documented feedback from the instructor regarding service boundaries.
- **Challenges:** Adapting to immediate feedback regarding tighter decoupling of the data logic layer.
- **Action Plans:** Re-factor Data Access Objects (DAO) to minimize interdependent system joins.

Week 10

Weekly Progress

- **Undertaken Tasks:**

- Expanded the prototype features towards Deliverable 3 completeness targets.
- Connected backend controller logic endpoints to the interface panels developed by Zihan.
- **Individual Contributions:**
 - Wrote core backend functional modules handling user authentications, login sessions and material downloads.
 - Exposed operational endpoints to pass information cleanly using structured parameters.
- **Challenges:** Correlating data model transfers cleanly with the asynchronous UI component triggers.
- **Action Plans:** Finalize full-flow prototype testing ahead of the Phase 3 target date.

GenAI Usage Record

- **Date:** Week 10
- **Tool Used:** Gemini
- **Purpose:** Debugging data marshalling mismatches between server modules.
- **Prompt Used:** Fix a Java runtime error where data from an inbound HTTP stream fails to parse correctly into an internal object representation due to timestamp mismatches.
- **Output Summary:** Identified date format string patterns and provided standard parse configuration properties.
- **How It Was Used:** Applied parsing fixes to ensure course deadlines display accurately.

Week 11 (PHASE 3 SUBMISSION)

Weekly Progress

- **Undertaken Tasks:**
 - Completed **Phase 3 (Deliverable 3): Prototype Progress.**

- Formally packaged codebases, pushed files to the remote GitHub repository and compiled verification screenshots.
- **Individual Contributions:**
 - Completed the system backend implementation details.
 - Verified all routing paths worked under test scenarios and gathered system logs for proof of operation.
- **Challenges:** Managing conflicting version merges across branches right before final compilation.
- **Action Plans:** Draft technical descriptions of the implemented codebase for the final report.

GenAI Usage Record

- **Date:** Week 11
- **Tool Used:** Gemini
- **Purpose:** Formatting markdown files for GitHub presentation.
- **Prompt Used:** Generate a clean README layout detailing core system run prerequisites, environment variables and initialization instructions for a cloud-connected application.
- **Output Summary:** Provided an organized Markdown template with clean code block structures and badge placeholders.
- **How It Was Used:** Adapted into the project's root GitHub documentation file.

Week 12

Weekly Progress

- **Undertaken Tasks:**
 - Initiated structural compilation for the final Phase 4 report.
- **Individual Contributions:**
 - Managed the full end-to-end execution path presentation during the observation.

- Drafted the definitive **Technology Stack** section for the final report, outlining our tools, cloud hosting environments and development frameworks.
- **Challenges:** Aligning runtime exceptions observed on alternative test platforms.
- **Action Plans:** Coordinate with Zihan to align testing outcomes with the architecture evaluation metrics.

GenAI Usage Record

- **Date:** Week 12
- **Tool Used:** Gemini
- **Purpose:** Formulating professional descriptions of technical frameworks.
- **Prompt Used:** Write an analytical overview of why Java web architecture combined with a cloud-native database engine forms an enterprise-grade stack for educational systems.
- **Output Summary:** Highlighted architectural parameters including thread safety, robust memory handling and structural reliability.
- **How It Was Used:** Integrated into the Technology Stack documentation block in the final report.

Week 13 (PHASE 4 SUBMISSION)

Weekly Progress

- **Undertaken Tasks:**
 - Completed, consolidated and **submitted Phase 4 (Deliverable 4): Comprehensive Project Report and Individual Logbook.**
- **Individual Contributions:**
 - Authored the comprehensive **Prototype Implementation** narrative, walking through component definitions and data connection processes.

- Collaborated with all team members to draft the project's collective **Conclusion and Reflection** section.
- Assembled and compiled this individual logbook matching personal contributions.
- **Challenges:** Ensuring uniform formatting across a large report document from three different authors.
- **Action Plans:** Finalize presentation materials for the upcoming Phase 5 Final Demonstration.

GenAI Usage Record

- **Date:** Week 13
- **Tool Used:** Gemini
- **Purpose:** Refinement and proofreading of the project conclusion.
- **Prompt Used:** Refine this concluding paragraph for a software architecture course report to sound professional, emphasizing architectural integrity and team collaboration.
- **Output Summary:** Provided a polished, academic closing summary focused on cloud-native robustness, theoretical modeling compliance and project success metrics.
- **How It Was Used:** Polished the final project report's shared closing text block.